

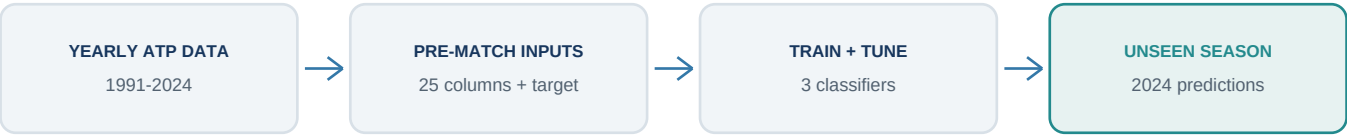
ATP Tennis Match Winner Prediction

A simple machine-learning study of pre-match information and ATP singles outcomes | 1991-2024 data

PROJECT QUESTION

Can a model predict whether one ATP player beats another using only information available before the match? We compare three classifiers, tune them on an earlier season, and test on a later season.

Project flow



Data and preparation

Source and scope	34 yearly ATP files (1991-2024); 108,375 match records; one combined Excel workbook.
Preparation	Standardize dates and numeric fields; check player IDs and duplicate match keys; preserve optional missing values for training-only imputation.
Model cohort	98,530 eligible model rows after excluding Davis Cup; the source workbook retains the combined match data for reference.

What one training example contains

One match is one row. Player A is consistently assigned as the lower numeric player ID; the target is 1 if A won and 0 if B won. The target is the answer used for learning, never an input. Player IDs and names are not model features.

Inputs are 25 pre-match columns, grouped as: (1) ranking and ranking points for both players plus their advantage; (2) age, height, and handedness differences; (3) prior career, last-5/10/25, and surface win-rate/count differences; (4) prior head-to-head and recent match activity; and (5) match context: surface, tournament level, round, best-of, and draw size.

Keeping the timeline honest

The source has tournament start dates rather than exact match dates. Prior event results enter player history only when that event started more than 14 days earlier; matches sharing a start date use the same history snapshot. Current-match winner, score, duration, aces, and serve statistics are excluded from inputs. Missing numeric values are imputed from training data; categories are one-hot encoded inside the model pipeline.

Training, tuning, and test results

The seasons are split in time order so the test represents a future season relative to model fitting.

Fit candidate models	Choose settings	Refit selected models	Final evaluation
1991-2021	2022 validation	1991-2023	2024 test

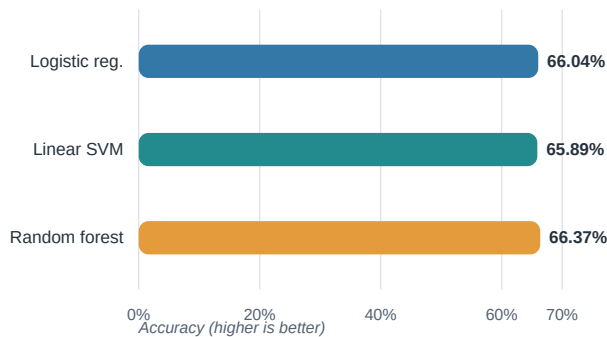
Algorithms and hyperparameters

Model	Settings tried	Selected on 2022
Logistic regression	$C = 0.01, 0.1, 1, 10$	$C = 0.01$
Linear SVM	$C = 0.01, 0.1, 1, 10$	$C = 0.01$
Random forest	Depth = None or 20; leaf = 1 or 3; 200 trees	Depth = 20; leaf = 3

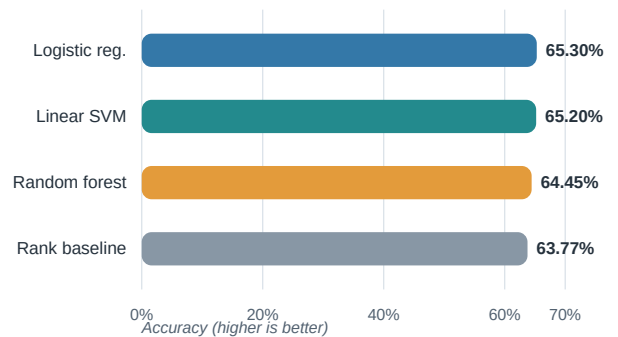
Each candidate is trained on 1991-2021 and selected by 2022 accuracy. Preprocessing is fitted only on each training set. The selected version is then refitted through 2023. Earlier 2024 events update history features for later events after the 14-day gap, but 2024 is not used to fit models or select settings.

Accuracy comparison

2022 validation



2024 held-out test



The rank baseline predicts the player with the better ATP ranking (lower rank number). It is available for 2,810 of the 2,816 test matches; the chart compares all four methods on those same 2,810 matches. Logistic regression scored 65.30% on that shared subset and 65.34% across all 2,816 matches. Accuracy is the share predicted correctly.

Takeaway and limits

The tuned models modestly exceed the ranking-only baseline in this test. The scores describe this 2024 evaluation; they do not guarantee future performance. The data lacks exact match dates, injuries, and betting-market information, and its ATP match coverage and missing fields limit what the models can learn.

Data attribution

ATP match data collected by Jeff Sackmann / Tennis Abstract; mirror: [farhadGithub/tennis-atp-data](https://github.com/farhad/tennis-atp-data). License: CC BY-NC-SA 4.0 (attribution, non-commercial use, share-alike).